

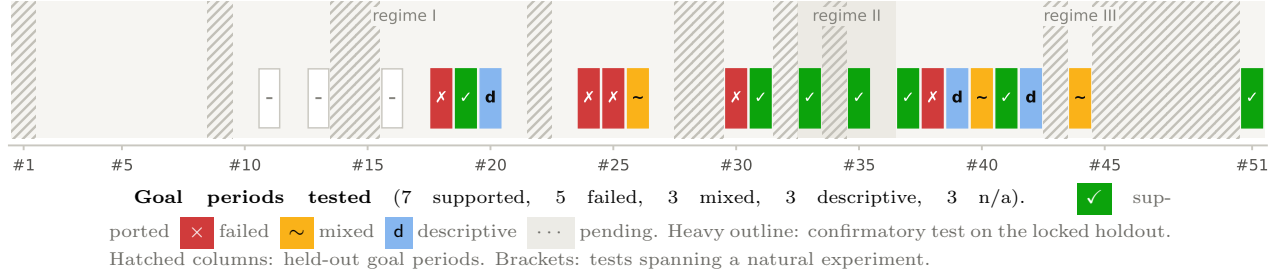
H11: Division of labor vs herding is the sign of a Potts coupling, set by the goal

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 2.5/5

Bottom line. LLM agents herd onto shared artifacts by default, in their commits as well as their attention, and spread out only when the task gives each one its own.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Per 30-min window and room, each agent's state is the project it touches most (strict artifact mentions). A uniform Curie–Weiss Potts model was fitted per week in 14 non-holdout weeks. Prediction: the sign of βJ follows the goal mode. **It failed:** $\beta J_{CW} > 0$ in 11/14 weeks whatever the mode (mode rule 5/11 signs, $p = 0.73$). Coupling beyond each agent's own preferences beats circular-shift and day nulls in 10/11 shared-artifact weeks; spread appears only in own-artifact weeks, as fixed specialization. **Round 1b** (deterministic labels, DQ4 work ledger, DQ6 ballots) changes no verdict. Agents' *work commits* herd too, in 4/6 shared weeks and onto the repos they attend (same repo in 84% of agent-windows), so “attention herds, work stays private” fails except in own-artifact and private-role weeks. On the correct #26 election the runoff jumps $1/3 \rightarrow 7/8$ in 83 s.

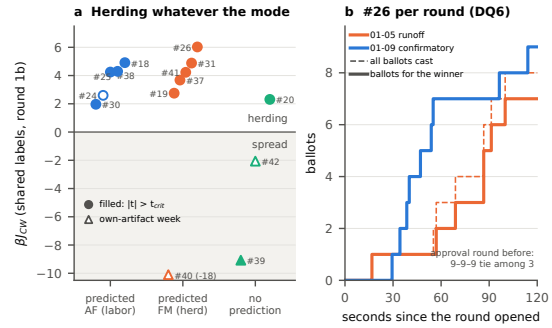
PHYSICAL MODEL

Model 10 (Potts), Curie–Weiss form. Spin $\sigma_i \in \{0, 1..q\}$: agent i 's project in a window ($q \leq 8$; 0 = rare projects, uncoupled), from artifact mentions (attention) or, in round 1b, from agent work commits (work); b a room block of N_b agents. h_a : popularity field of project a ; βJ : one uniform coupling (> 0 herding, < 0 division of labor). βJ_c , βJ_s : mean-field first-order point and ordered spinodal. A variant with agent fields $h_i(a)$ separates coupling from specialization.

$$P(\sigma) \propto \exp \left[\sum_i h_{\sigma_i} + \frac{\beta J}{N_b - 1} \sum_{i < j \in b} \delta_{\sigma_i \sigma_j} [\sigma_i \geq 1] \right]$$

$$x_a = \text{softmax}_a(h_a + \beta J x_a), \quad x_1 : \frac{1}{q} \rightarrow \frac{q-1}{q} \text{ at } \beta J_c$$

$$\beta J_c = \frac{2(q-1)\ln(q-1)}{q-2}, \quad \beta J_s = \min_s \frac{1}{s} \ln \frac{1+(q-1)s}{1-s}$$



Observables. (a) βJ_{CW} per tested week on the round-1b shared labels, grouped by the class predicted from the goal text: every predicted division-of-labor week (blue) herds; negative values only in own-artifact weeks (triangles), reproduced by a within-agent permutation. #40 clipped (-18). (b) #26 on DQ6 ballots: after a 9–9–9 approval tie, DeepSeek-V3.2 takes 7 of 8 runoff ballots within 100 s (orange) and 9 of 9 in the 01-09 confirmatory vote (blue).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The default is herding, and it is real work, not just looking. In shared-artifact weeks agents commit to the same repos in the same half-hour, beyond their stable preferences, and their attention labels point at the repos they commit to. Division of labor appeared only when the task handed each agent its own artifact or role; there, agents still look at each other's work more than they contribute to it. Design rule (still post hoc): assign ownership if you want parallel work. The coupling-beyond-fields z on artifact links is a cheap duplicated-work monitor that now tracks commits. The Potts machinery bought little: no clean phase transition, drive and coupling not separated, and the one jump (#26) is a voting step that looks like a ballot cascade.

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RESULTS VS PREDICTIONS

Prediction	Observed (round 1 → 1b)	Verdict
P1 sign of βJ_{CW} follows mode	5/11 signs, $p = 0.73$ (unchanged); same sign in 14/14	mixed by rule; failed
P2 coupling beyond fields by mode	AF weeks $z_{N2} > 0$ in 5/5; #38 failed +2.1 → +1.9	failed
P3 first-order consensus	0/4 project jumps; #26 vote failed; jump 1/3 → 7/8 (correct round), at threshold βJ 3.2 [1.05, 6.0] vs 2.75	failed; jump
P5 held-out gain	8/10 → 10/10	supported
R1b-1 replication	14/14 P1, 12/14 P2 identical	supported
R1b-2a work herds (shared)	$z_{N2} \geq 2$ in 4/6	supported
HH266 work private	work $z < 2$ in 2/5 herding not weeks	ported
#26 field, not cascade	runoff voters saw 2.6 earlier ballots; field favoured Gemini	failed
#40 hub less worked	hub 0.73 of work vs 0.74 of attention	failed

GOAL PERIODS IN DETAIL

G18 × 1b: P1 failed (sig.); P2 failed (sig.) (unchanged)
G19 ✓ 1b: P1 supported; P2 supported (unchanged)
G20 d 1b: descriptive (unchanged)
G24 × 1b: P1 failed; P2 failed (sig.) (unchanged)
G25 × 1b: P1 failed (sig.); P2 failed (sig.) (unchanged)
G26 ~ 1b: mixed (native: jump on the right vote; cascade, n...
G30 × 1b: P1 failed (sig.); P2 failed (sig.) (unchanged)
G31 ✓ 1b: supported (native: work herds)
G33 ✓ 1b: supported (work herds)
G35 ✓ 1b: supported (work herds)
+11 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

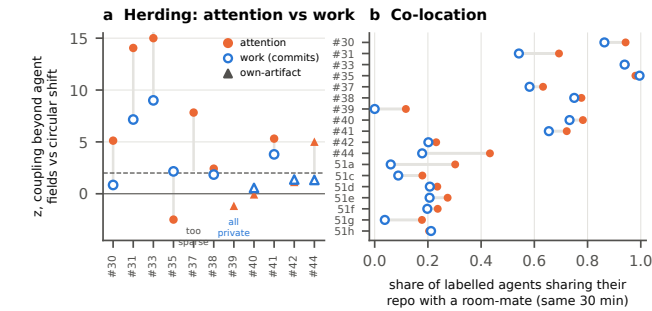
CAVEATS

Drive and coupling are still not separable: “herding” is an effective coupling. z values near 2 move by up to 0.5 between null draws (#20, #38). Work is public agent commits only (dense from #30; author time). In #51 the work-space pseudo-likelihood is ill-determined (each agent has its own repo), so co-location is the measure there. #35, a shared week not used in round 1, has $z_{N2} = -2.5$ in attention, which would refute the frozen confirmatory clause C2; disclose before the holdout run. The #26 cascade reading rests on 8 ballots and no text. The ownership rule remains post hoc.

ROUND 2

A kinetic Potts with ledger read-outs for ballots and commits (does seeing k earlier ballots or commits raise adoption?); file-level states inside shared repos to look for division of labor below the repo; NE15 as an intervention.

- **Essence.** Why do agents pile on: imitation, or attraction to where the work already is?
- **H11-R1** Preferential attachment: $P(\text{join } X) \propto (\text{recent activity on } X)^\alpha$.
- **H11-R2** Herding is stigmergic, not social: artifact activity predicts joining better than chat mentions of X (E3, vs H28).
- **H11-R3** Herding is a coordination solution that raises output; own-artifact spread duplicates effort (test against the work ledger).



Second figure. (a) Coupling beyond agent fields (z vs circular shift) in attention (filled) and work (open), #30–#44: work herds where attention does, except #30 and #38; own-artifact weeks (triangles) stay below 2. (b) Co-location (agents sharing their repo with a room-mate in the same 30 min): equal in shared weeks, lower in work in #44 and every #51 unit.

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